

Unit 2

Virtualization

Virtualization is a technology that uses software to create virtual versions of physical computing resources, such as servers, storage, networks, and operating systems.

It allows multiple virtual machines (VMs) to run on a single physical machine, enhancing hardware utilization and enabling efficient resource allocation.

Key benefits include cost reduction, improved scalability, increased energy efficiency, and better disaster recovery, making it a foundational technology for cloud computing.

➤ VIRTUALIZATION REFERENCE MODEL

Virtualization Reference Model (VRM) is a conceptual framework that describes how virtualization components interact in a cloud computing environment.

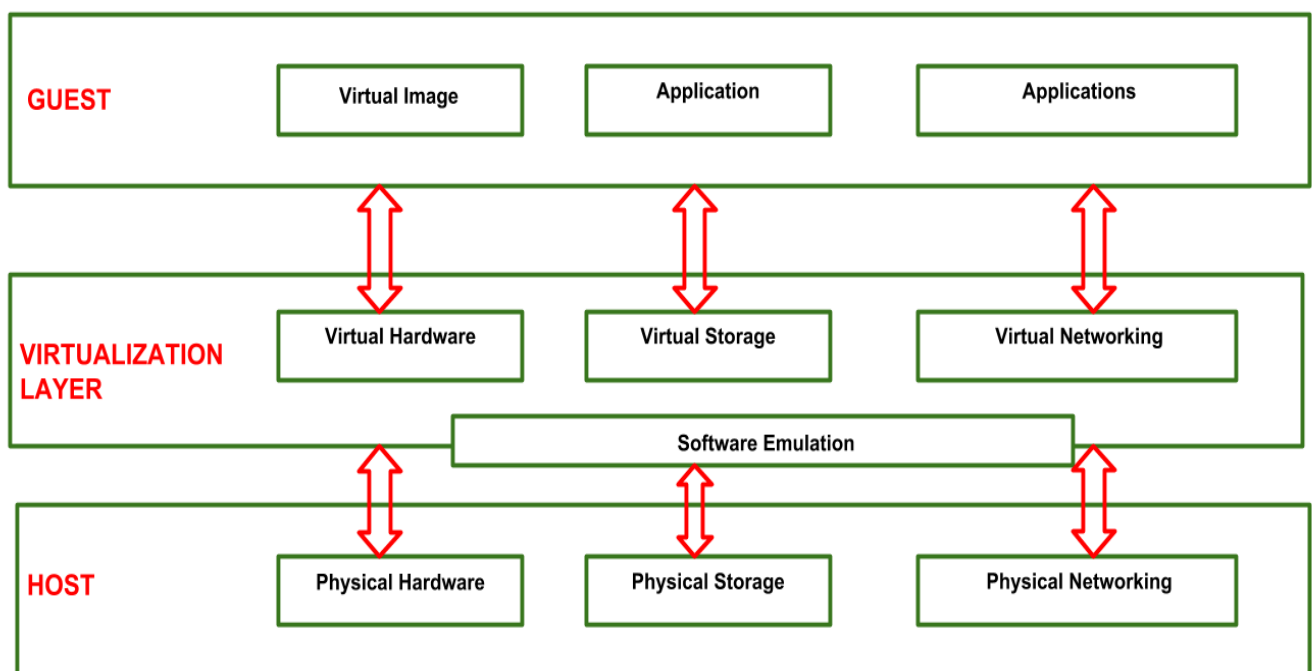


Figure: Reference Model of Virtualization.

There are the main layers of VRM :-

1. HOST

This is the physical foundation, consists of the **actual hardware** like **processors, memory, storage, and network devices**.

The **operating system** on the host is often responsible for managing these physical resources.

2. Virtualization Layer (Hypervisor):

Hypervisor is also called as **Virtual Machine Monitor**.

This is the **core** of the model, a **software** layer that **creates** and **manages virtual machines**.

It **abstracts** the **physical resources**, presenting a **virtualized version** to the guests.

The hypervisor intercepts requests from the guests and translates them for the physical hardware.

The virtualization layer provides the guest with a virtualized version of the hardware, giving the guest the impression that it has exclusive access to a complete system.

Types of hypervisors:

Type 1 Hypervisor (Bare-Metal Hypervisor):

The hypervisor is **installed directly** on the underlying host system.

It does **not require** any **base server operating system**.

It has direct access to **hardware resources**.

Examples :

VMware ESXi, Citrix XenServer, and Microsoft Hyper-V hypervisor.

Type 2 Hypervisor:

It is run over an **installed operating system** (such as Windows or macOS).

Examples :

VMware Workstation, Oracle VirtualBox.

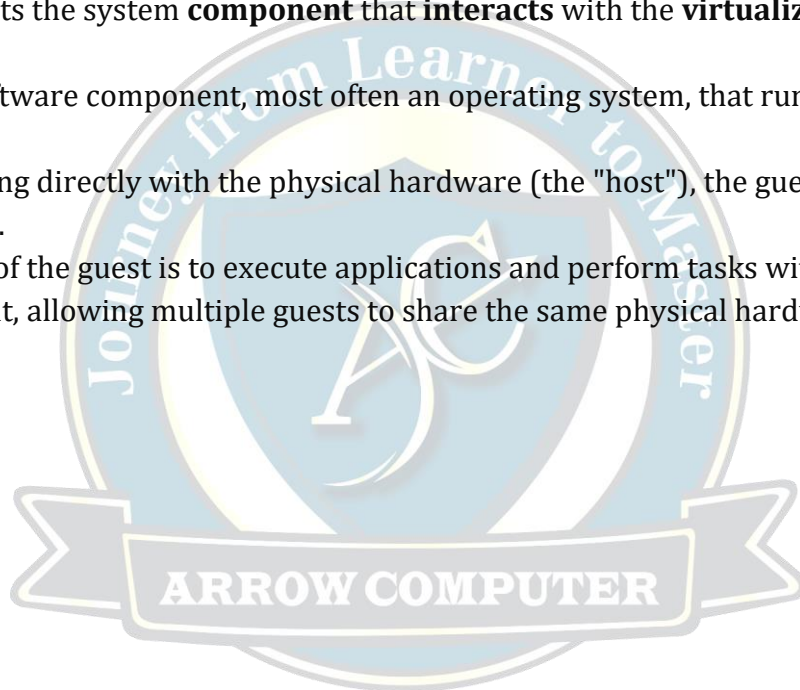
3. GUEST

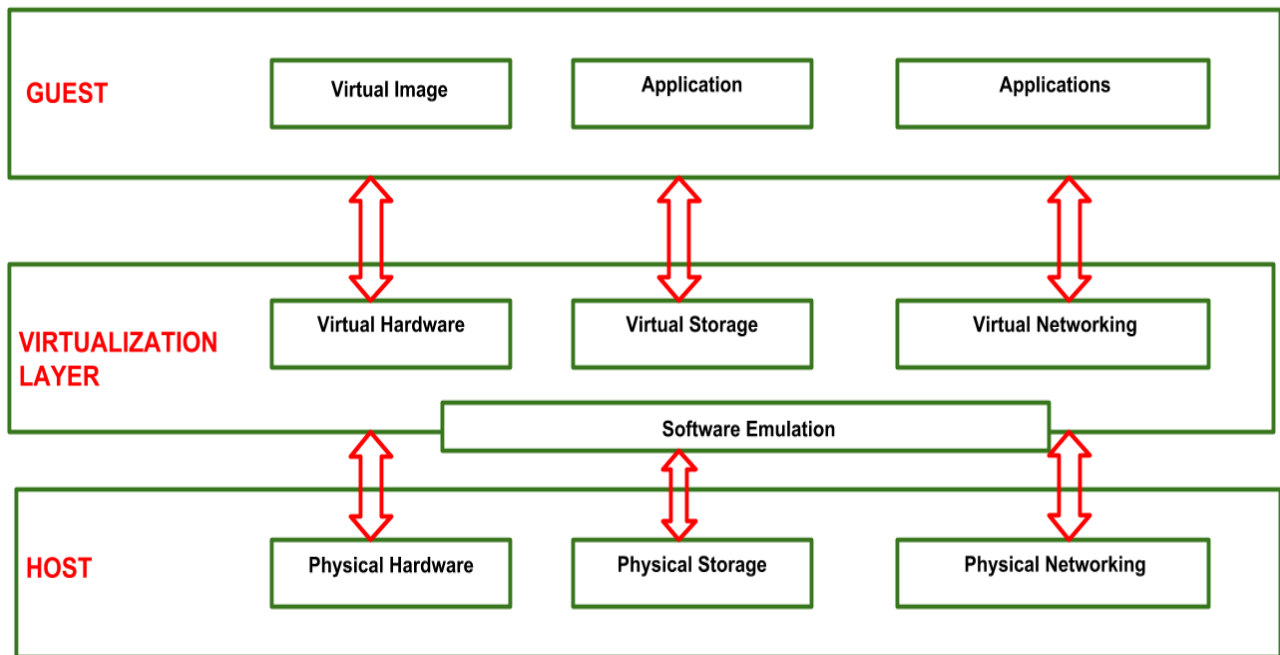
The guest represents the system **component** that **interacts** with the **virtualization layer** rather than with the **host**.

The guest is the software component, most often an operating system, that runs within the virtual environment.

Instead of interacting directly with the physical hardware (the "host"), the guest interacts with the virtualization layer.

The main purpose of the guest is to execute applications and perform tasks within its isolated virtual environment, allowing multiple guests to share the same physical hardware.





Characteristics Of Virtualization

Hardware Abstraction & Resource Pooling

Virtualization abstracts the underlying physical hardware, allowing IT managers to create a virtual layer over physical resources like servers, storage, and networks.

These pooled resources can then be dynamically allocated to virtual machines (VMs) as needed.

Sharing

Multiple virtual machines or computing environments can share the same host physical machine and its resources (CPU, memory, storage). This improves hardware utilization and reduces the number of physical servers required, leading to cost and energy savings.

Isolation of Resources

Each virtual machine is isolated from the others, and they cannot access each other's resources. This provides security and stability, as a problem with one virtual machine will not affect the others.

Emulation

Emulation means using a **program** or a **device** to **imitate** the **working** of **another program** or **device**.

Flexibility and Scalability

Virtual machines can be easily created, modified, and destroyed, providing significant flexibility to adapt to changing demands. Resources can be scaled up or down quickly without needing new physical hardware, a concept known as elasticity.

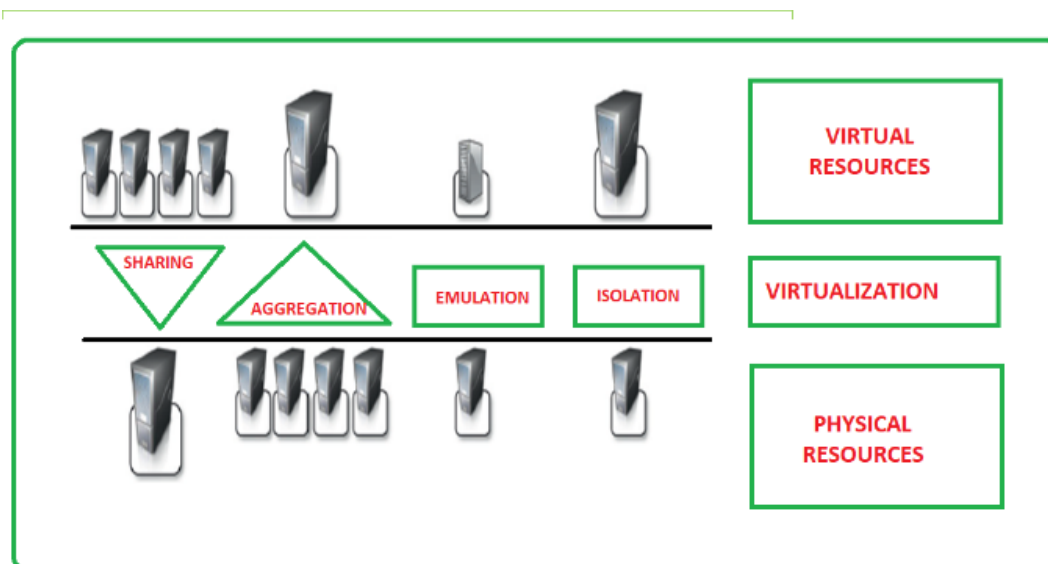
Portability

Virtualization also provides portability, as virtual machines can be easily moved between physical machines. This allows for easy disaster recovery, as virtual machines can be quickly moved to a different physical machine in the event of a disaster. It also allows for easy migration between

physical machines, as virtual machines can be moved to new hardware without affecting the applications or data.

Aggregation

In contrast to sharing, aggregation combines **multiple separate physical hosts** into a single, unified virtual resource that appears as one large server to the user.



Types of Virtualization

1. Storage Virtualization

Storage virtualization refers to the process of **combining storage** from **multiple physical storage** devices (like **hard drives, SSDs, and storage arrays**) into a **single, unified, logical storage resource**.

This technology **abstracts** away the **complexities** of the **underlying hardware**, **simplifying management, enhancing resource utilization**, and providing greater **flexibility** and data **mobility**.

It creates a **virtual layer** between the **user** and the **physical storage**, allowing for on-demand addition and removal of storage, load balancing, and centralized management.

How it Works

1. Pooling Physical Storage:

Various physical storage devices (HDDs, SSDs, storage arrays) are combined into a single logical storage pool.

2. Abstraction Layer:

A virtual layer is created between the host system (applications) and the physical storage hardware.

3. Virtual Resources:

This layer presents a single, unified "virtual" storage unit, masking the underlying hardware details.

Benefits of Storage Virtualization:

- **Centralized Management:**

A single, unified management console, often called a single pane of glass, to monitor and maintain multiple virtualized storage arrays reduces the time and effort necessary to manage the physical systems individually.

This is particularly beneficial when a large number of storage systems or storage systems from multiple vendors are in the virtualization pool.

- **Better Storage Utilization:**

Physically disparate storage devices are pooled together, making it easier to allocate and utilize the available capacity more efficiently, reducing wasted space.

- **Cost Reduction:**

Virtual storage requires fewer hardware devices and software licenses than traditional enterprise storage architectures.

This can save organizations a significant amount of money.

Furthermore, virtualization supports dynamic storage provisioning, offering a more scalable and cost-effective way to add storage as the organization's needs change.

- **Improved Performance:**

Workload distribution and parallel processing of data requests can significantly improve overall system performance and reduce latency.

- **Less downtime risk Easier Data Migration**

Virtualized environments provide fault tolerance, allowing the migration of data and applications from one server to another with minimal downtime. Also, virtual redundancy reduces the risk of disruption, increases storage flexibility, improves storage performance and reduces failure risk.

- **Extended life of older storage systems**

Virtualization can extend the usefulness of older storage hardware by incorporating them into the pool as a tier for less critical data.

- **Enhanced Scalability:**

Organizations can dynamically expand their storage by adding resources to the virtual pool, even across different physical devices, without disrupting operations.

- **Dynamic Flexibility:**

Virtual volumes can be resized, moved, or replicated, to meet changing demands, allowing for seamless scaling and adaptation to evolving business needs.

- **High Availability:**

Features such as data replication, mirroring, and automated failover ensure high uptime and uninterrupted access to data, even if a component fails.

- **Simplified Disaster Recovery:**

Features like automated backups and snapshots make data protection and recovery processes more effective.

2. Network Virtualization:

Network virtualization is a fundamental technology in cloud computing that enables providers to create multiple logical, or virtual, networks that run on a single physical network infrastructure.

It abstracts network resources like switches and routers, decoupling them from the physical hardware and delivering them as software.

This process allows for the flexible and efficient provision of network services in cloud environments.

Example :- Virtual LAN, SDN (Software Defined Network)

Advantages :-

- **Flexibility and Agility:**
- **Efficiency and Cost Savings:**
- **Simplified Management:**
- **Enhanced Security:**
- **Rapid Service Delivery:**

3. Desktop Virtualization:

Desktop virtualization is a process in which you can create different virtual desktops that users can use from any device like laptop, tablet.

It's great for users who need flexibility, as it simplifies software updates and provides portability.

How it Works

1. Separation:

The desktop operating system, applications, and user data are detached from the physical hardware.

2. Centralized Hosting:

These virtual desktops are hosted on a powerful server or within a cloud infrastructure.

3. Access:

Users connect to their virtual desktop using a client device, such as a laptop, tablet, or thin client, over a network connection.

4. Processing:

The virtual desktop performs all processing on the server, rather than the user's local device.

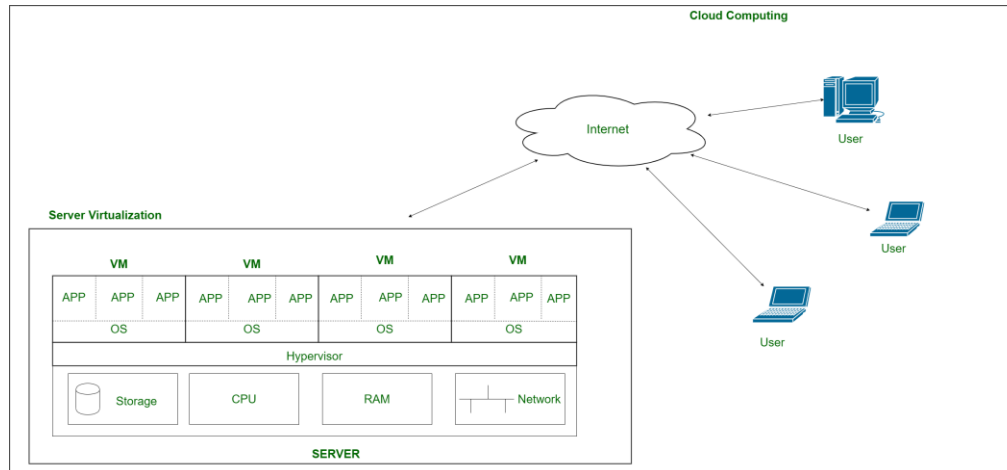
Advantages:-

- Anywhere, Anytime Access
- Centralized Management
- Enhanced Security
- Improved Agility
- Device Flexibility

4. Server virtualization

Server virtualization is a technology that creates multiple independent virtual servers on a single physical server by separating server software from the underlying hardware. This is achieved through a hypervisor, which allocates CPU, memory, storage, and network resources to virtual machines (VMs).

Key benefits include improved resource utilization, reduced costs, greater flexibility, and faster deployment, making it the foundation for cloud computing.



5. Application Virtualization

Application virtualization is a technology that runs applications on a remote server and streams them to users' devices, making them appear as if they are installed locally, without the need for a full installation on the end-user's machine.

This method separates applications and their dependencies from the underlying operating system, allowing for more flexible and efficient management, deployment, and access to software from virtually any device.

6. Data virtualization

Data virtualization is a method of data integration that creates a single, unified view of data from diverse sources without needing to physically move or store the data in a central repository. It uses a virtual, abstraction layer to query and combine data from multiple locations in real-time, providing a single point of access for users and applications while hiding the technical complexities of the underlying sources.

➤ VMWare: Full Virtualization Reference Model

VMware is a leading provider of virtualization technology, and its full virtualization solution allows multiple operating systems to run on a single physical machine. This is accomplished through a hypervisor, which sits between the hardware and the operating systems.

Full virtualization is a technique that allows multiple operating systems (OS) to run on a single physical computer, making each OS think it has its own hardware.

This is done without needing to change or modify the guest operating systems.

VMware is a popular example of this technology.

How it works

VMware's approach to full virtualization, particularly on older x86 architecture before hardware-assisted technology became common, involves a combination of two techniques:

- **Direct execution:** The hypervisor allows non-critical instructions from the guest OS to run directly on the host CPU for high performance.
- **Binary translation:** For sensitive or privileged instructions that could affect the system, the hypervisor intercepts and translates them. It replaces the sensitive instructions with new, safe instruction sequences that have the same intended effect on the virtual hardware.

VMware full virtualization reference model components

1. Physical Hardware

This is the base layer consisting of the physical server, including the x86 CPU, memory (RAM), storage (disks), and network interface cards (NICs). All virtual resources are abstracted from this physical infrastructure.

2. Hypervisor (Virtual Machine Monitor)

The hypervisor is the core software layer that sits directly on the physical hardware in a Type 1 architecture (e.g., ESXi) or on top of a host OS in a Type 2 architecture (e.g., Workstation).

It is responsible for managing the hardware and allocating resources to the virtual machines.

It provides the virtualization layer that decouples the VMs from the underlying hardware.

3. Virtual Machines (VMs)

Each virtual machine is a software-based, emulated version of a physical computer.

- **Encapsulation:** A VM encapsulates all the components required to run a full operating system, including the virtual CPU, virtual RAM, virtual BIOS, and virtual disks.

4. Guest Operating System

The guest OS is the operating system running inside the virtual machine. In full virtualization, it runs without any modifications because it believes it is operating on a physical machine.

5. Virtual Devices

To maintain the illusion of a physical machine for the guest OS, the hypervisor provides virtual representations of hardware devices. These include:

- **Virtual network adapters (vNICs):** For network connectivity.
- **Virtual disk controllers:** For accessing virtual disks.

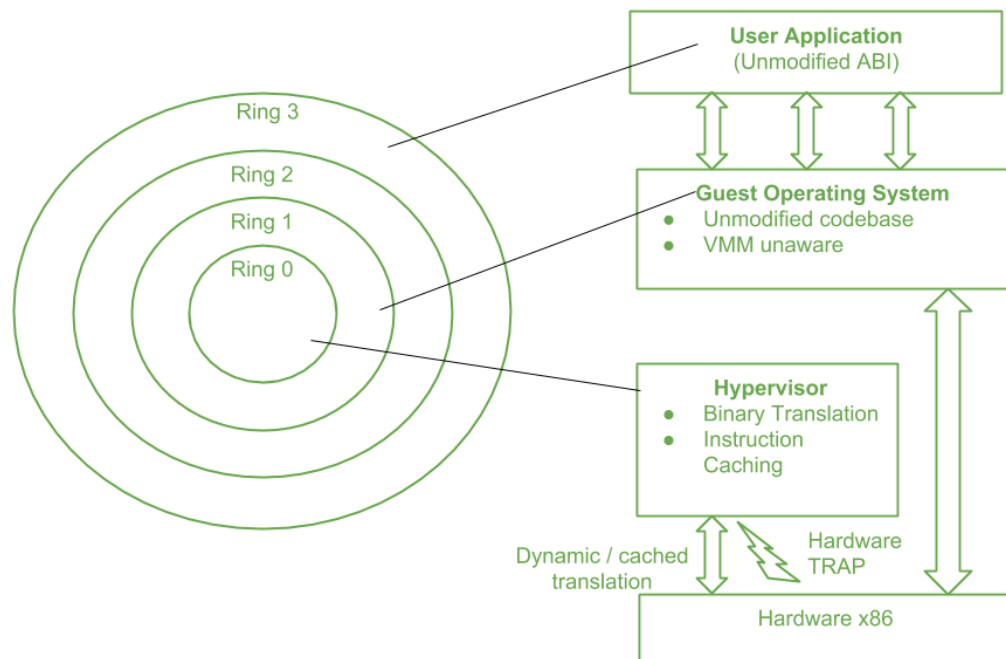


Figure - Full Virtualization Reference Model